

Text Recognition & Extraction From the Argyle Theatre Collection



Research Rationale & Purpose

Objective:

Digitise and structure data from the *Argyle Theatre Collection (c.1890–1930)*

Rationale:

1. Historical theatre records are valuable primary sources
2. Currently stored as scans, limiting searchability and access
3. Manual cataloguing restrict large scale analysis

Purpose:

Enable automated data extraction to improve accessibility and support research



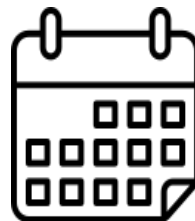
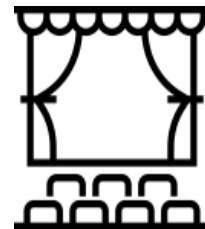
Key Outputs

Develop an automated system to extract:



01 Performer Names

02 Types of Acts

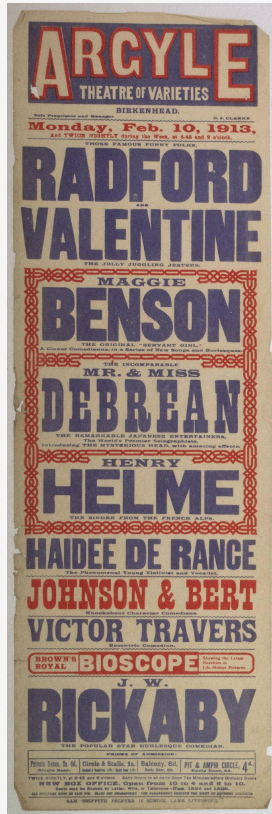


03 Dates of Performance

Methodology: LLM Image Extraction

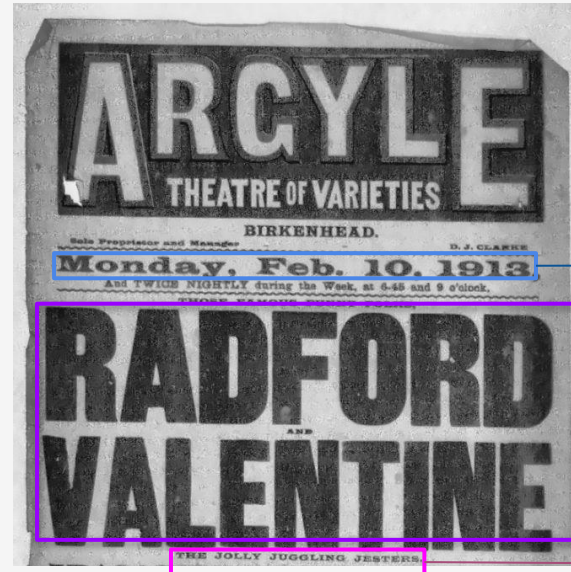


Model Name	Gemini (2.5 Pro)	Qwen (2.5-VL in 8-bit quantisation)
<i>Data Preparation</i>	☐ Deskewing ☐ Denoising ☐ Contrast Limited Adaptive Histogram Equalization (CLAHE) enhancement	☐ Deskewing ☐ Denoising ☐ CLAHE enhancement ☐ Adaptive Binarisation ☐ Multiscale Tiling
<i>Prompt Framework</i>	Persona Task Context Format (PTCF)	
<i>Data Extraction</i>	Transcribed into structured metadata and format (in CSV and JSON) while preserving their original artistic phrasing	



(NFA178R12-205.jpg)

Data
Preparation



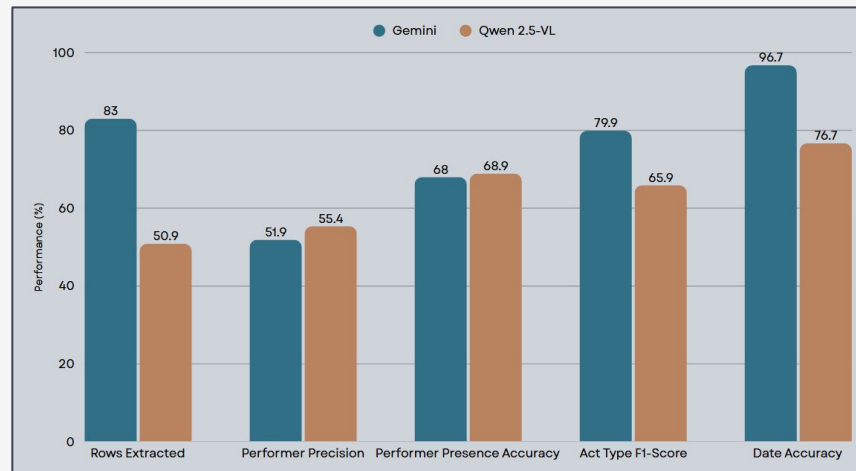
accession_id	performance_date	performer	act_name	act_type	image_source
178R12	Monday, Feb 10, 1913	RADFORD AND VALENTINE	THE JOLLY JUGGLING JESTERS.	Jugglers	NFA178R12-205.jpg

Results Comparison



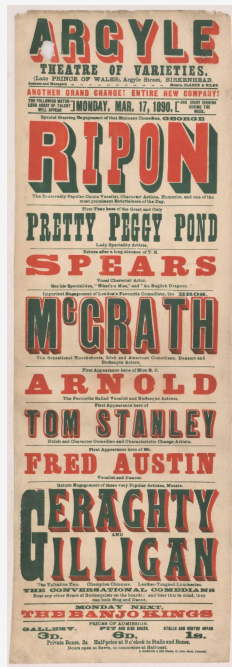
Model Name	Gemini (2.5 Pro)	Qwen (2.5-VL in 8-bit quantisation)
Rows Extracted (out of the original 283)	235 (83.0%)	144 (50.9%)
Performer Names Precision	51.9%	55.4%
Performer Names Presence Accuracy	68.0%	68.9%
Act Type Weighted F1-Score	79.9%	65.9%
Date of Performance Accuracy	96.7%	76.7%

Model performance comparison across different metrics



Unsuccessful Extraction Justification

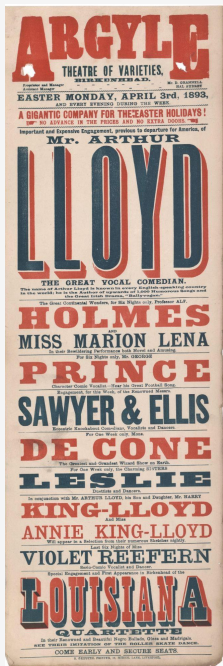
Rejected



(NFA178R12-10.jpg)

VS.

Successful

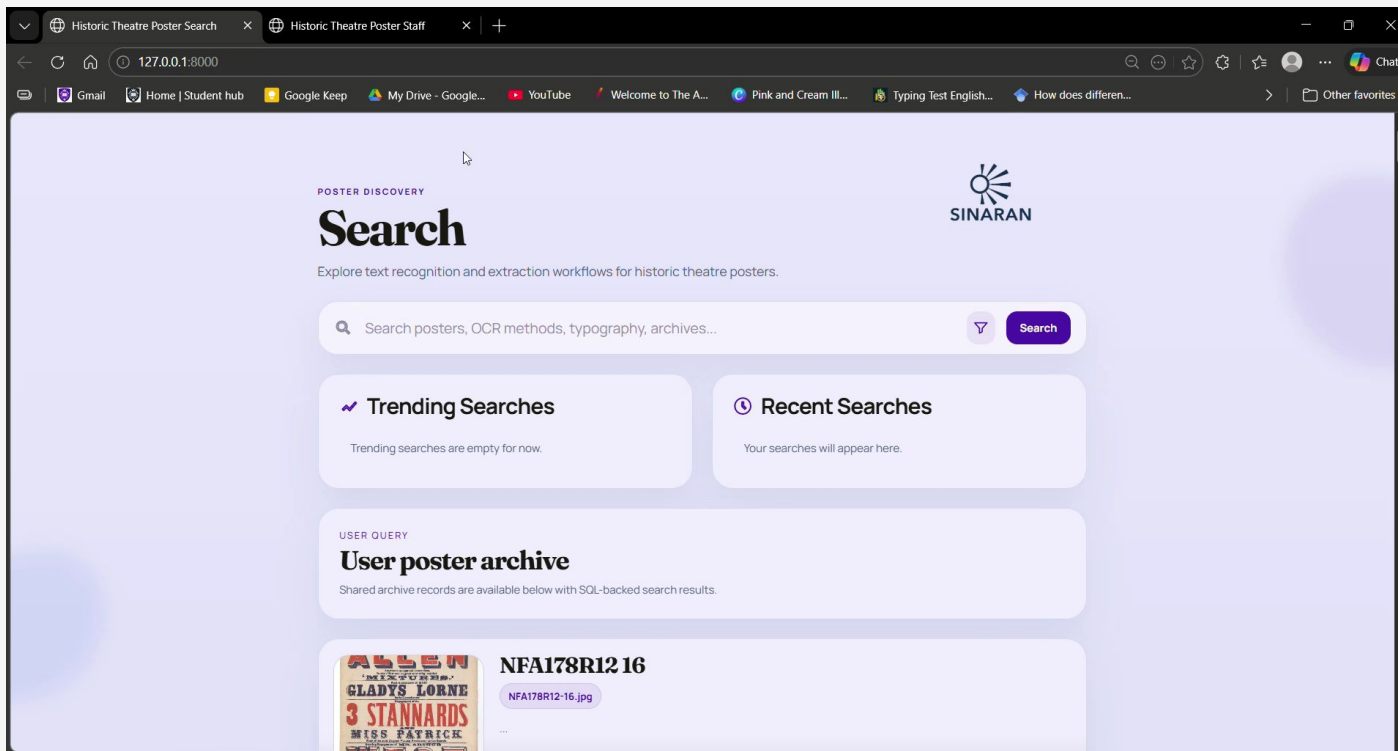


(NFA178R12-38.jpg)

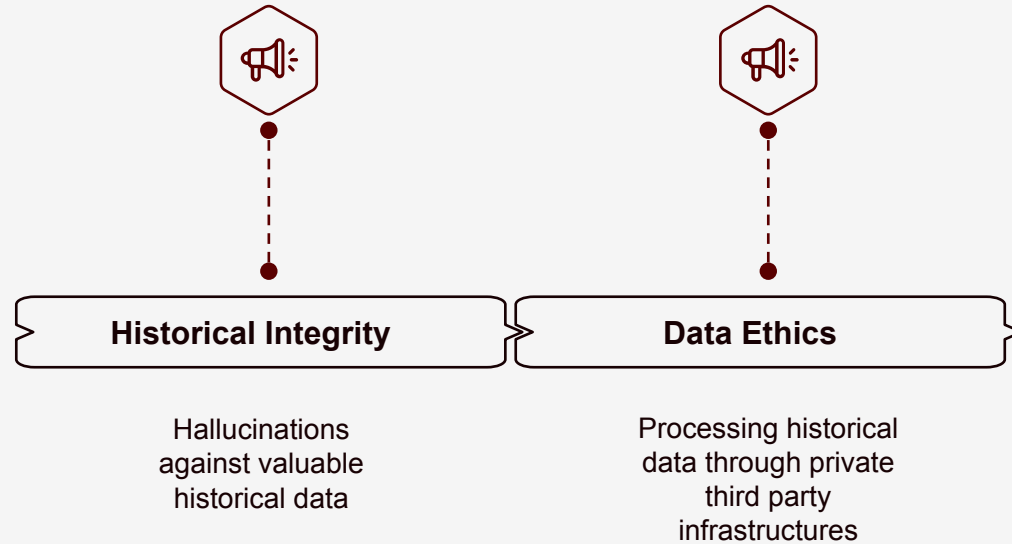
 Gemini

	Blur Score (Laplacian)	Instances (out of 30)
Successful	674	24
Rejected	347	6
Difference	94.1%	20%

Implementation: Website PoC



Ethical Implications





SINARAN

Thank You!

Declarations (AI & IBM)



AI supports us during initial brainstorming phases and debugging codes during development process. We ensured no sensitive data were processed and adhered to ethical guidelines



Design thinking helped us oriented our decision making and final product towards our users for example challenging any assumption made by users when navigating our solution